



Question Bank

Institute Name:	Silver Oak College of Computer Application	Programme Name:	Master of Science Data Science
Semester:	1	Academic Year:	2026-2027
Course Name:	Information and Data Security	Course Code:	DSM5061C

Sr. No.	Question	Marks	Cognitive Level of Bloom's Taxonomy	CO No.
Unit 1: Foundations of Information and Data Security				
1	Define Information Security and Data Security. State any two objectives of information security.	2	Remember	CO1
2	Explain any three types of information assets in a data-driven organization.	3	Understand	CO1
3	Explain the Confidentiality, Integrity and Availability (CIA) Triad with one suitable example for each principle.	4	Understand	CO1
4	Differentiate between threats, vulnerabilities and risks. Explain their relationship with the attack surface using a suitable example.	5	Analyze	CO1
5	Explain the role of risk assessment in preventing data breaches. Mention the major steps involved in a basic risk assessment process.	5	Analyze	CO1
6	What is social engineering? Explain phishing and any two human-based attack techniques.	4	Understand	CO1
7	Differentiate between identity theft and insider threats. Give one example of each.	3	Analyze	CO1
8	Differentiate between authentication and authorization. Explain how access control supports information security.	4	Analyze	CO1
9	Explain password security practices and the role of Multi-Factor Authentication (MFA) in protecting user accounts.	5	Understand	CO1



10	Differentiate between symmetric and asymmetric encryption with respect to keys, speed and applications.	5	Analyze	CO1
11	Explain digital signatures and describe how they provide authentication, integrity and non-repudiation in cryptographic applications.	5	Analyze	CO1
Unit 2: Data Privacy and Protection				
12	Define data privacy and state any two privacy principles.	2	Remember	CO2
13	Differentiate between Personal Data, Sensitive Personal Data and Personally Identifiable Information (PII) with examples.	3	Understand	CO2
14	Explain the major privacy requirements that should be considered during data collection and secure data acquisition.	4	Understand	CO2
15	Explain secure data storage and data management practices used to protect organizational information.	4	Understand	CO2
16	Explain the security and privacy considerations involved in data processing and data sharing.	5	Analyze	CO2
17	Describe secure data disposal and data destruction techniques. Why is secure destruction important for privacy protection?	4	Understand	CO2
18	Explain data classification and data categorization. How do they help organizations implement appropriate security controls?	5	Analyze	CO2
19	Explain data masking and any three data protection techniques used to safeguard sensitive information.	5	Understand	CO2
20	Differentiate between data anonymization and pseudonymization with suitable examples.	4	Analyze	CO2
21	Explain the best practices for secure file sharing and secure data storage in an organization.	3	Understand	CO2
22	Explain data backup, data recovery and business continuity. Describe their importance during a data loss or system failure.	5	Analyze	CO2
23	What is a digital footprint? Explain any three practices for protecting privacy on social media platforms.	3	Understand	CO2



24	Analyze the major data privacy challenges faced when collecting, processing and sharing data for digital analytics.	5	Analyze	CO2
25	Explain the role of data protection policies in establishing secure data handling practices within an organization.	4	Understand	CO2
26	Explain data handling best practices that employees should follow while accessing, storing, processing and sharing organizational data.	5	Apply	CO2
27	What is Data Loss Prevention (DLP)? Explain how DLP awareness helps prevent accidental or unauthorized data leakage.	5	Analyze	CO2
Unit 3: Security Technologies for Data-Driven Systems				
28	Define network security and state any two network security mechanisms.	2	Remember	CO3
29	Explain the role of firewalls in protecting network traffic. Mention any two firewall functions.	3	Understand	CO3
30	Explain VPN and describe how it provides secure communication over an untrusted network.	4	Understand	CO3
31	Explain cloud computing and discuss the major security challenges associated with cloud environments.	5	Analyze	CO3
32	Explain the Cloud Shared Responsibility Model with suitable examples of responsibilities of cloud providers and customers.	5	Understand	CO3
33	Explain the major security challenges associated with Big Data environments.	4	Understand	CO3
34	Explain data access control in Big Data systems. How can appropriate access controls protect sensitive data?	5	Analyze	CO3
35	Explain the major data security challenges in distributed systems and suggest suitable security measures.	5	Analyze	CO3
36	What is Artificial Intelligence in cybersecurity? Explain any two applications of AI for improving cybersecurity.	3	Understand	CO3



37	Explain AI-assisted threat detection and security automation with suitable examples.	4	Apply	CO3
38	Explain the major security risks associated with AI systems and machine learning models.	5	Analyze	CO3
39	What is data poisoning? Explain its impact on machine learning models with a suitable example.	3	Understand	CO3
40	Explain adversarial attacks and distinguish them from data poisoning attacks.	4	Analyze	CO3
41	Explain deepfakes and Generative AI security risks. Discuss their potential impact on individuals and organizations.	5	Analyze	CO3
42	Analyze any three emerging trends in network, cloud, Big Data or AI security.	5	Analyze	CO3
43	Discuss how AI, cloud computing and automation are shaping the future of cybersecurity.	5	Evaluate	CO3
Unit 4: Data Governance, Compliance and Ethics				
46	Define data governance and state any two reasons for its importance in an organization.	2	Remember	CO4
47	Differentiate between data ownership and data stewardship with suitable examples.	3	Understand	CO4
48	Explain the major components of a data governance framework and their role in organizational data management.	4	Understand	CO4
49	Explain the key provisions of the Information Technology Act, 2000 related to information security and cyber activities.	5	Understand	CO4
50	Explain the objectives and major principles of the Digital Personal Data Protection (DPDP) Act, 2023.	5	Understand	CO4
51	Explain the importance of data privacy regulations and compliance requirements for organizations handling personal data.	4	Analyze	CO4
52	Explain consent management and discuss its role in achieving privacy compliance during personal data processing.	5	Analyze	CO4



53	Explain responsible data usage and ethical data collection practices with suitable examples.	4	Apply	CO4
54	Explain ethical AI practices. Discuss the importance of bias mitigation and fairness in AI systems.	5	Analyze	CO4
55	Define security governance and explain the role of security policies in an organization.	3	Understand	CO4
56	Explain the major steps involved in risk management and discuss their importance in security governance.	5	Analyze	CO4
57	Explain the role of security standards and compliance frameworks in establishing effective information security practices.	4	Understand	CO4
58	Explain ISO/IEC 27001 and describe the purpose of an Information Security Management System (ISMS).	5	Understand	CO4
59	Explain the NIST Cybersecurity Framework (NIST CSF) and its major functions for managing cybersecurity risks.	5	Understand	CO4
60	Evaluate how data governance, security governance, regulatory compliance and ethical practices can be integrated to build a responsible data-driven organization.	5	Evaluate	CO4

Course Coordinator

HoD / HoI